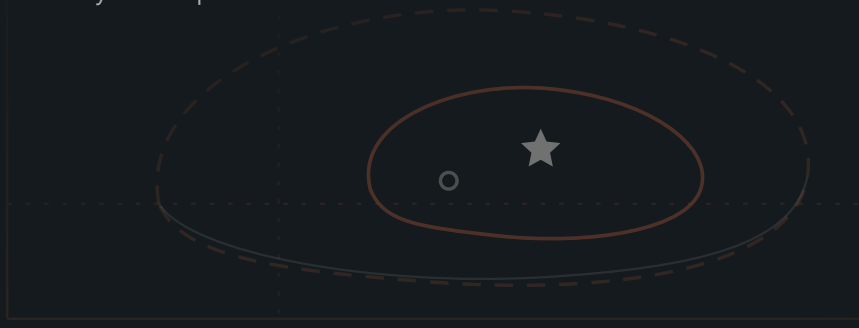
REPRODUCIBILITY PACKAGE

VERSION 1.1.0 · 28 OCTOBER 2026

zeroleak

Permeability from porosity in three phases, with the phase boundaries fixed in advance and never fitted.

Five public bodies, one of which ships with the package. One power law per porosity phase, six free parameters, and a baseline that does not read the answer it is scored against. **The package reproduces every number the paper prints**, down to the last digit, from the archives or from the one body that ships with it.



The surface behind the fixed pair. Worst-body loss over all 951 evaluated cutpoint pairs; the star marks the minimax pair (0.110, 0.200), the open circle the pair that ships (0.10, 0.20). The two differ by 0.0009 in R^2.

BODIES

SPE 10 Tarbert · Upper Ness
Volve 15/9-19A
Brent 211/29 · Piper 15/17

CONTENTS

Source · tests
reference outputs
80 tests · 951 cells

LICENCE

MIT (code)
CC BY 4.0 (data)

DOI

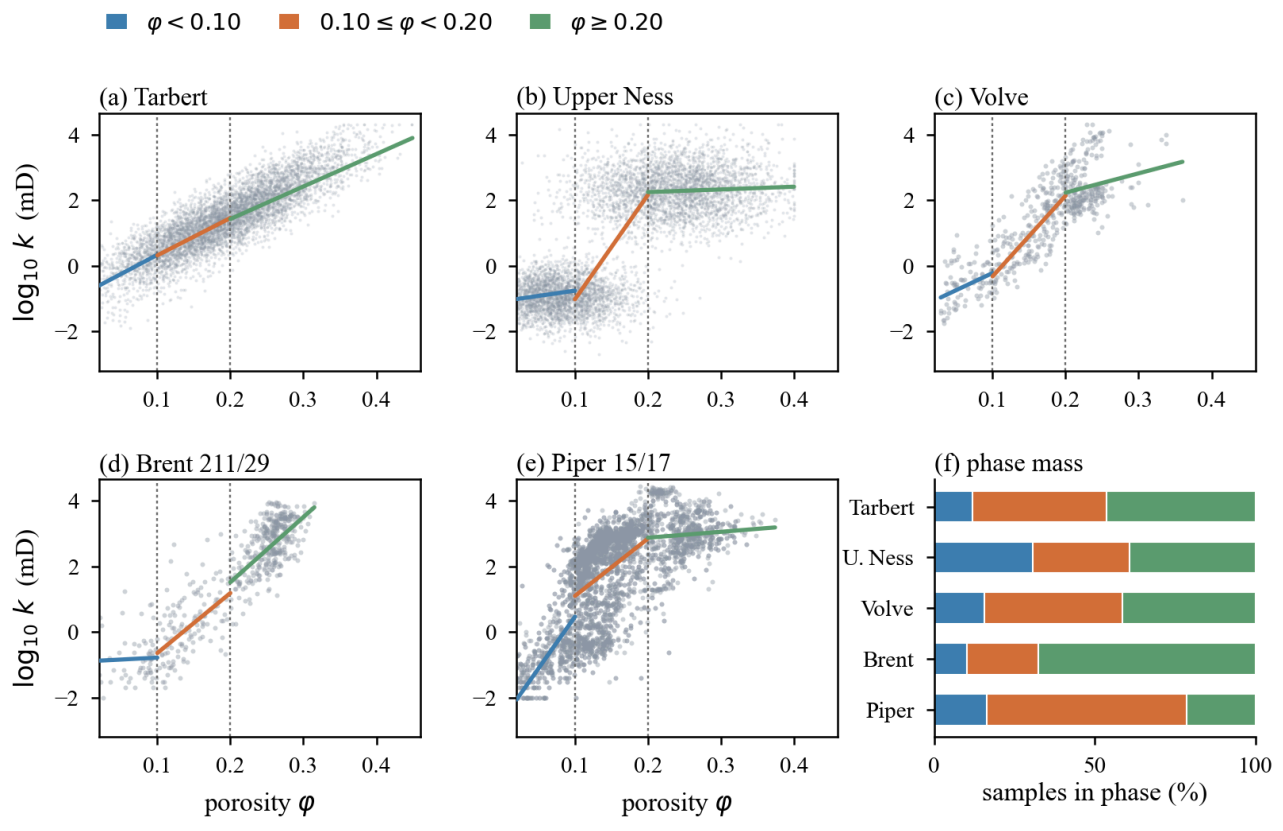
10.5281/zenodo.XXXXXXX
to be minted

A cutpoint you never fit, and a baseline that never reads the answer.

Most permeability correlations are scored against a baseline built from the very quantity being predicted. The usual one is the flow zone index, and it cannot be computed without k.

The flow zone index is `RQI / ϕz` with `RQI = 0.0314 √(k / ϕ)`. It is a genuinely useful grouping variable, and it is also a function of the target. Any accuracy advantage it shows over a porosity-only model is therefore not evidence about prediction: the information was supplied rather than inferred. Here the baseline sees porosity and nothing else, and the three phase boundaries sit at fixed porosities — 0.10 and 0.20 — chosen before any body is scored and never re-optimised inside the folds.

What remains is a deliberately small model: three phases, one power law in each, six free parameters, fitted by ordinary least squares in log space. Its job is not to win a leaderboard. It exists so that a claim about transferability can be stated without the target leaking into the thing that measures it.



Five bodies, one protocol. Panels (a)–(e): core-plug porosity against permeability, split at the fixed porosities φ = 0.10 and 0.20 (dotted), with the per-phase power law fitted inside each band. Panel (f): how the samples distribute across the three phases — equal-frequency in the paper's own parameterisation, and visibly uneven in φ on the multi-well fields.

NOT A CLASSIFIER

No hydraulic flow units are assigned, and no rock type is named. The phases are porosity intervals, nothing more.

NOT A RANKING CLAIM

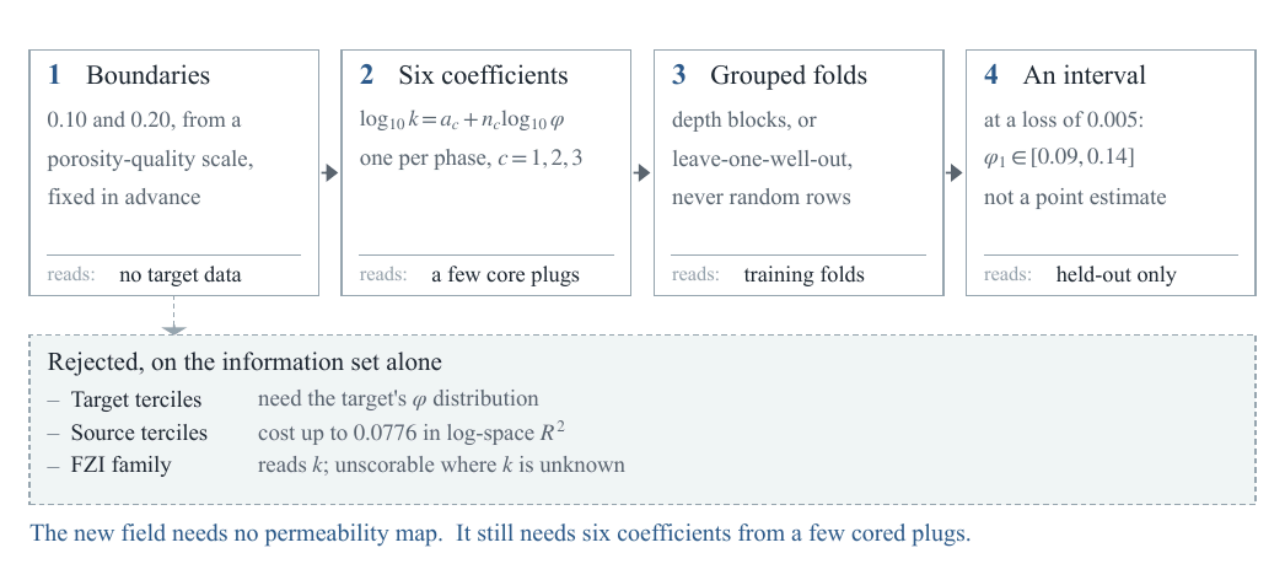
R^2 in k space is reported as a diagnostic. It is not used to order the five bodies against each other.

NOT A SUBSTITUTE

Nothing here replaces core description. The model says what porosity alone carries, and stops there.

Four decisions, each with a stated information set.

The model is small because the protocol forbids it from being larger. Each step below names what it is allowed to read, and the last panel names the alternatives that were rejected on that basis alone.



Bounds, coefficients, folds, an interval. The bottom panel is the part that matters: target terciles need the target's own porosity distribution, source terciles cost up to 0.0776 in log-space R^2, and the flow-zone-index family reads k, which is unobservable where k is unknown.

Two consequences follow, and they are worth stating plainly because both are easy to read as a weakness. First, **the cutpoints are not estimated**: they are fixed at 0.10 and 0.20 from a porosity-quality scale, before any body is scored, so no fold can tune them. Second, **the contribution is a range, not a point**: at a tolerance of 0.005 in loss, the admissible lower cutpoint spans $\varphi_1 \in [0.09, 0.14]$. A paper that reported a single optimum would be claiming a precision this design cannot defend.

The last line of the figure is the one a practitioner cares about: a new field needs no permeability map, but it does need six coefficients from a few cored plugs. That is the whole deployment story, and it is deliberately modest.

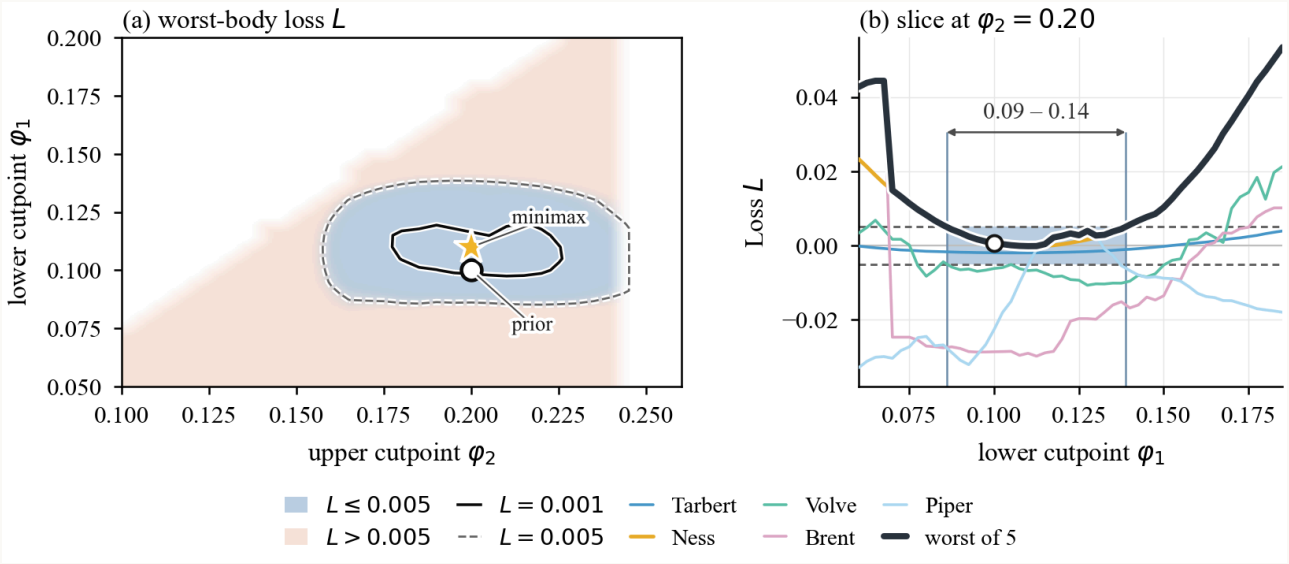
> *Nothing here is fitted that could have been fixed, and nothing is fixed that the data could have decided.*
>
> THE DESIGN RULE, IN ONE SENTENCE

Six coefficients per body, and the surface behind the cutpoints.

LEAVE-ONE-FOLD-OUT $R^2(\log_{10} K)$, SHIPPED CONFIGURATION

Body	n	Prior 0.10/0.20	Terciles fitted	RMSE / σ
SPE 10 Tarbert	100 000	0.7355	0.7336	0.5143
SPE 10 Upper Ness	100 000	0.6634	0.6641	0.5802
Volve 15/9-19A	557	0.7207	0.7145	0.5285
Brent 211/29	586	0.7430	0.7143	0.5070
Piper 15/17	4 428	0.4868	0.4642	0.7164

The two grid bodies are scored on a fixed 100 000-point subsample. Columns three and four differ by at most 0.029 — fitting the boundaries buys almost nothing, so fixing them costs almost nothing.



Why the pair survives being questioned. Left: worst-body loss over all 951 evaluated pairs. Right: a slice at φ_2 = 0.20, with the ε = 0.005 band shaded — the admissible window is an interval, and the reported pair sits inside it rather than at a sharp optimum.

951

pairs searched, K fixed at 3, six parameters throughout

0.110 · 0.200

where the worst body is least bad (L = −0.00018)

+0.0009

what the fixed pair gives up there, in R^2

Every number in the paper has a command behind it.

Three claims are load-bearing, and each is one run away. None of them requires the full archives.

10 / 10

Table values reproduced to four decimals, five bodies × two strategies, in a three-second run.

0.00e+00

Largest disagreement with the archived twenty-draw set, across 100 individual losses.

557

Predictions compared cell by cell by `verify`, not a summary statistic standing in for them.

The body that ships with the package

Volve 15/9-19A comes with the code: 557 core plugs, full input, so the whole chain runs on a clean machine with no download and no credentials. The other four are other people's data and need the archives described in `DATA.md`; none of them is redistributed here.

```
$ python -m zeroleak verify --config configs/base.yaml
# 557 predictions scored first, then one line per reference file
file                                          status  detail
index.csv                                     ok
volve_15_9_19a__prior.config.json             ok
volve_15_9_19a__prior.metrics.csv             ok
volve_15_9_19a__prior.pred.csv                ok
... three more, then
verify: all files match expected/
```

> **What the check refuses to claim.** Brent 211/29 and Piper 15/17 are folded by well, so there is no boundary to redraw; they are reported as *fixed* rather than as a standard deviation of zero. The check speaks for three bodies out of five, and says so in its own output.

Spread of the loss at the reported pair

Twenty draws, each with its own subsample seed and fold offset. The offsets are listed in the source rather than generated, because the sentence “the boundaries move by 1–59 cells” admits many sequences of twenty points, and only one of them produced the printed table.

Body	Baseline	Mean	SD	Min	Max
SPE 10 Tarbert	−0.00206	−0.00183	0.00026	−0.00224	−0.00124
SPE 10 Upper Ness	+0.00032	+0.00056	0.00015	+0.00024	+0.00079
Volve 15/9-19A	−0.00630	−0.00632	0.00272	−0.01328	−0.00201
Brent · Piper	−0.02863 · −0.02262	unchanged	fixed	unchanged	unchanged

Four commands, and a check that can fail.

```
# Python 3.11 · numpy 2.4.4 · nothing below needs the network
# run from inside zeroleak-1.1.0/, the directory the archive unpacks to
pip install -e ".[test]"

# reproduces the table from the body that ships with the package
python -m zeroleak run    --config configs/base.yaml
# the same, then a cell-by-cell diff against the reference output
python -m zeroleak verify --config configs/base.yaml

# all five bodies (~5 s), the 951-cell scan (~3.5 min), twenty draws (~25 s)
export ZEROLEAK_DATA_DIR=/path/to/archives   # layout in DATA.md
python -m zeroleak run     --config configs/full.yaml
python -m zeroleak scan    --config configs/full.yaml
python -m zeroleak noise   --config configs/full.yaml
python -m pytest                             # 80 passed
```

What is in the archive

`zeroleak-1.1.0.zip`	Source, tests, both configurations, the shipped body, the reference outputs and this manual. Unzips to a directory that installs with one command.
`zeroleak-1.1.0-manual.pdf`	This document — the version to read before unzipping anything.
`zeroleak-1.1.0-README.md`	The same instructions as plain text, for reading in the browser without downloading anything.

Why the tests are worth something

Eighty of them, and the interesting ones are the ones that must fail: the verifier is handed an altered reference and required to exit non-zero, and the loss sign is reversed to confirm a named assertion breaks. A checker that can only print ok moves the burden of proof onto the reader without giving them anything to push against.

CITE THIS PACKAGE

```
@software{zeroleak_2026,
  title     = {zeroleak: permeability from porosity in three phases,
               with fixed cutpoints},
  version   = {1.1.0},
  year      = {2026},
  publisher = {Zenodo},
  doi       = {10.5281/zenodo.XXXXXXX}
}
```